

Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ MI/E	First Test - Modeule E	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **6 Marks**

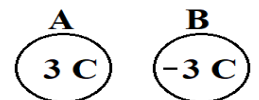
1) The electric flux increases as the area increases		
2) A suck pressure is a negative gauge pressure		
3) In Pascal principle, the force does not change.		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) Which of the following has positive charge:

a) Electron	b) Proton	c) Neutron	d) Electron & Proton	e) None of these
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2) Two spheres shown below. The charge on sphere B is:

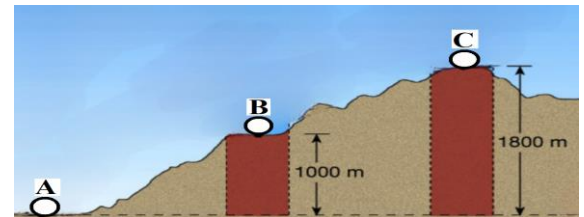


a) 1.5 C	b) 0 C	c) - 3 C	d) 3 C	e) None of these
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3) The force 2 N. If a distance changes from 8 m into 2 m. The force is:

a) 32 N	b) 0.5 N	c) 16 N	d) 2 N	e) None of these
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4) From the figure shown below. The pressures



a) $P_B > P_A$	b) $P_C < P_B$	c) $P_A = P_B$	d) $P_B < P_C$	e) None of these
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5) A Fully immersed object in fluid will be at rest if:

a) $\rho_o < \rho_f$	b) $\rho_o > \rho_f$	c) $\rho_f > \rho_o$	d) $\rho_o = \rho_f$	e) None of these
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6) 25 N object displaces 10 N of water. The weight of object in water is:

a) 10 N	b) 25 N	c) 15 N	d) 0 N	e) None of these
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7) Object with $4\text{ m} \times 2\text{ m} \times 3\text{ m}$ having a density of $x\text{ kg/m}^3$. The pressure in Pa on the ground is:

a) $1.25 x$	b) $0.125 x$	c) $3 x$	d) $30 x$	e) None of these
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Section III: Answer all the following by selecting the correct choice

Full and clear steps are required

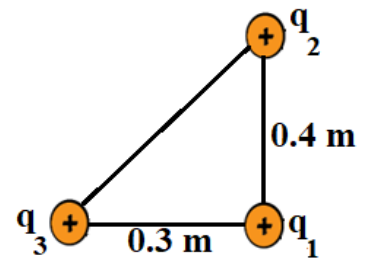
20 Marks

Constant		
Electron: $m = 9.11 \times 10^{-31}\text{ kg}$	$K_e = 9 \times 10^9\text{ N.m}^2/\text{C}^2$	Gravity: $g = 10\text{ m/s}^2$
Proton: $m = 1.67 \times 10^{-27}\text{ kg}$	$G = 6.67 \times 10^{-11}\text{ N.m}^2/\text{kg}^2$	Water: $\rho_f = 1000\text{ kg/m}^3$

1) 400 N object in air and 120 N when it is fully immersed in a fluid of $\rho_f = 1500\text{ kg/m}^3$. The volume of the metal is:

a) 0.0187 m^3	b) 0.0337 m^3	c) 0.008 m^3	d) 0.0147 m^3	e) None of these
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2) The total electric force on q_3 is: $q_1 = 2 \mu\text{C}$, $q_2 = 1 \mu\text{C}$ and $q_3 = 3 \mu\text{C}$



a) 0.67 N	b) 0.37 N	c) 0.75 N	d) 0.13 N	e) None of these
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3) Two opposite charges one is 4 times the other separated by distance r , if a ratio of charge to a distance r is $2 \mu\text{C}/\text{m}$. The electric force on $3q$ placed at distance r from a point where $E = 0$ is:

a) 1.11 N	b) 0 N	c) 7.5 N	d) 0.63 N	e) None of these
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4) A room has dimensions of 4 m and 2 m and a height of 3 m. If the air density is $1.21 \text{ kg}/\text{m}^3$. The weight of the air in the room is:

a) 217.8 N	b) 290.4 N	c) 320.7 N	d) 435.6 N	e) None of these
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5) Two objects having $q_1 = 20 \mu\text{C}$ and $q_2 = 10 \mu\text{C}$, separated by 30 cm. The electrostatic force is:

a) 6 N	b) 20 N	c) $0.002 \times 10^{-3} \text{ N}$	d) $2.22 \times 10^{-15} \text{ N}$	e) None of these
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